
DSC5001: Statistical Machine Learning I

Tutorial 1: Probability, Bayes, and Uncertainty

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Tutorial Setup

Format

All tasks use reasoning and written calculation. No software is required.

- ▶ **Task 1:** Work with a discrete probability model.
- ▶ **Task 2:** Perform Bayesian reasoning with multiple sources of uncertainty.
- ▶ **Task 3:** Combine two uncertain measurements using variance and covariance.
- ▶ **Optional Challenge:** Examine why zero covariance does not imply independence.

Give a numerical answer where appropriate, but justify the statistical reasoning behind it.

Task 1: A Discrete Probability Model

Task type: Probability / Calculation / Modeling

Suggested time: 12 minutes

Let X denote the number of high-priority alerts received by a server during a short monitoring interval. Suppose

$$X \in \{0, 1, 2, 3\}, \quad \mathbb{P}(X = x) = c(x + 1), \quad x = 0, 1, 2, 3.$$

1. Determine the normalizing constant c .
2. Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.
3. Compute $\mathbb{P}(X \geq 2 \mid X \geq 1)$.
4. For alert counts over a longer fixed time interval, which model family is most natural among Bernoulli, Binomial, Poisson, and Gaussian? State one assumption required for your choice.

Solution – Task 1: Normalize and Compute Moments

Since probabilities must sum to one,

$$c(1 + 2 + 3 + 4) = 10c = 1, \quad \boxed{c = 0.1}.$$

Hence,

$$\mathbb{P}(X = 0) = 0.1, \quad \mathbb{P}(X = 1) = 0.2, \quad \mathbb{P}(X = 2) = 0.3, \quad \mathbb{P}(X = 3) = 0.4.$$

$$\mathbb{E}[X] = 0(0.1) + 1(0.2) + 2(0.3) + 3(0.4) = \boxed{2}.$$

$$\mathbb{E}[X^2] = 0^2(0.1) + 1^2(0.2) + 2^2(0.3) + 3^2(0.4) = 5.$$

$$\boxed{\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 5 - 2^2 = 1.}$$

Solution – Task 1: Conditional Probability and Modeling

Since $\{X \geq 2\} \subseteq \{X \geq 1\}$,

$$\mathbb{P}(X \geq 2 \mid X \geq 1) = \frac{\mathbb{P}(X \geq 2)}{\mathbb{P}(X \geq 1)} = \frac{0.3 + 0.4}{0.2 + 0.3 + 0.4} = \boxed{\frac{7}{9} \approx 0.778}.$$

A natural starting point for event counts in a fixed time interval is a **Poisson distribution**.

Typical assumptions:

- ▶ events occur independently;
- ▶ events occur at an approximately constant average rate;
- ▶ multiple events in a very short interval are unlikely.

Common mistake

The given four-point distribution is not itself Poisson; the question asks for a plausible count model in a broader setting.

Task 2: Defects, Suppliers, and Inspection

Task type: Bayes / Conditional Probability

Suggested time: 15 minutes

A manufacturer obtains components from two suppliers:

$$\mathbb{P}(A) = 0.70, \quad \mathbb{P}(B) = 0.30.$$

Their defect rates are

$$\mathbb{P}(D \mid A) = 0.01, \quad \mathbb{P}(D \mid B) = 0.05.$$

An inspection system satisfies

$$\mathbb{P}(F \mid D) = 0.90, \quad \mathbb{P}(F \mid D^c) = 0.04.$$

Assume inspection performance is the same for both suppliers. Here D means defective and F means flagged.

1. Compute $\mathbb{P}(D)$.
2. Compute $\mathbb{P}(F)$.
3. Compute $\mathbb{P}(D \mid F)$.
4. Compute $\mathbb{P}(B \mid F)$.
5. Explain what observing F changes in Parts 3 and 4.

Solution – Task 2: Overall Probabilities

By the law of total probability,

$$\mathbb{P}(D) = \mathbb{P}(D \mid A)\mathbb{P}(A) + \mathbb{P}(D \mid B)\mathbb{P}(B).$$

$$\mathbb{P}(D) = 0.01(0.70) + 0.05(0.30) = \boxed{0.022}.$$

Therefore,

$$\mathbb{P}(D^c) = 0.978.$$

The overall probability of a flag is

$$\mathbb{P}(F) = \mathbb{P}(F \mid D)\mathbb{P}(D) + \mathbb{P}(F \mid D^c)\mathbb{P}(D^c).$$

$$\mathbb{P}(F) = 0.90(0.022) + 0.04(0.978) = \boxed{0.05892}.$$

Solution – Task 2: Defect after a Flag

Bayes' rule gives

$$\mathbb{P}(D \mid F) = \frac{\mathbb{P}(F \mid D)\mathbb{P}(D)}{\mathbb{P}(F)}.$$

Substitute the values:

$$\mathbb{P}(D \mid F) = \frac{0.90(0.022)}{0.05892} \approx \boxed{0.336}.$$

Thus,

$$\boxed{\mathbb{P}(D \mid F) \approx 33.6\%}.$$

Common mistake

The 90% inspection sensitivity is $\mathbb{P}(F \mid D)$, not $\mathbb{P}(D \mid F)$.

Solution – Task 2: Which Supplier?

First compute the probability of a flag given Supplier B:

$$\mathbb{P}(F \mid B) = \mathbb{P}(F \mid D, B)\mathbb{P}(D \mid B) + \mathbb{P}(F \mid D^c, B)\mathbb{P}(D^c \mid B).$$

Using the equal-performance assumption,

$$\mathbb{P}(F \mid B) = 0.90(0.05) + 0.04(0.95) = 0.083.$$

Bayes' rule gives

$$\mathbb{P}(B \mid F) = \frac{\mathbb{P}(F \mid B)\mathbb{P}(B)}{\mathbb{P}(F)} = \frac{0.083(0.30)}{0.05892} \approx \boxed{0.423}.$$

The flag increases the probability of Supplier B from 30% to about 42.3%.

Task 2: Additional Information

Task type: Bayes / Interpretation

Suggested time: 5 minutes

Recall that Supplier B has

$$\mathbb{P}(D \mid B) = 0.05,$$

and the inspection system satisfies

$$\mathbb{P}(F \mid D) = 0.90, \quad \mathbb{P}(F \mid D^c) = 0.04.$$

A component has been **flagged**, and you additionally learn that it came from **Supplier B**.

1. Before calculating, predict whether $\mathbb{P}(D \mid F, B)$ should be larger or smaller than $\mathbb{P}(D \mid F) \approx 0.336$.
2. Explain your prediction.
3. Calculate $\mathbb{P}(D \mid F, B)$.
4. Explain why learning the supplier changes the posterior.

Solution – Task 2: Additional Information

For Supplier B,

$$\mathbb{P}(F \mid B) = 0.90(0.05) + 0.04(0.95) = 0.083.$$

Therefore,

$$\mathbb{P}(D \mid F, B) = \frac{\mathbb{P}(F \mid D, B)\mathbb{P}(D \mid B)}{\mathbb{P}(F \mid B)}.$$

Hence,

$$\mathbb{P}(D \mid F, B) = \frac{0.90(0.05)}{0.083} \approx \boxed{0.542}.$$

Thus,

$$\boxed{\mathbb{P}(D \mid F, B) \approx 54.2\%}.$$

Since Supplier B has a higher defect rate,

$$54.2\% > 33.6\%.$$

Task 3: Combining Two Noisy Sensors

Task type: Expectation / Variance / Covariance

Suggested time: 18 minutes

Two sensors X and Y measure the same unknown quantity θ . Both are unbiased:

$$\mathbb{E}[X] = \mathbb{E}[Y] = \theta.$$

Their uncertainty is

$$\text{Var}(X) = 4, \quad \text{Var}(Y) = 9, \quad \text{Cov}(X, Y) = 3.$$

Consider

$$Z_w = wX + (1 - w)Y, \quad 0 \leq w \leq 1.$$

You may use

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

1. Show that Z_w is unbiased.
2. Derive $\text{Var}(Z_w)$.
3. Find the w minimizing $\text{Var}(Z_w)$.
4. Compare the minimum with $\text{Var}(X)$ and $\text{Var}(Y)$.
5. If $\text{Cov}(X, Y) = 0$, would combining them become more or less useful?

Solution – Task 3: Unbiasedness

By linearity of expectation,

$$\mathbb{E}[Z_w] = \mathbb{E}[wX + (1 - w)Y] = w\mathbb{E}[X] + (1 - w)\mathbb{E}[Y].$$

Since $\mathbb{E}[X] = \mathbb{E}[Y] = \theta$,

$$\mathbb{E}[Z_w] = w\theta + (1 - w)\theta = \boxed{\theta}.$$

Therefore, Z_w is unbiased for every w .

Solution – Task 3: Variance

Using

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y),$$

with $a = w$ and $b = 1 - w$,

$$\text{Var}(Z_w) = 4w^2 + 9(1 - w)^2 + 6w(1 - w).$$

Expanding,

$$\text{Var}(Z_w) = 4w^2 + 9 - 18w + 9w^2 + 6w - 6w^2.$$

Therefore,

$\text{Var}(Z_w) = 7w^2 - 12w + 9.$

Solution – Task 3: Optimal Weight

Minimize

$$\text{Var}(Z_w) = 7w^2 - 12w + 9.$$

Differentiate:

$$\frac{d}{dw} \text{Var}(Z_w) = 14w - 12.$$

Set the derivative equal to zero:

$$14w - 12 = 0.$$

Hence,

$$w^* = \frac{6}{7}.$$

The optimal estimator is

$$Z_{w^*} = \frac{6}{7}X + \frac{1}{7}Y.$$

Solution – Task 3: Minimum Variance

Substitute $w^* = 6/7$ into

$$\text{Var}(Z_w) = 7w^2 - 12w + 9.$$

Then

$$\text{Var}(Z_{w^*}) = 7\left(\frac{6}{7}\right)^2 - 12\left(\frac{6}{7}\right) + 9 = \boxed{\frac{27}{7} \approx 3.86}.$$

Since

$$3.86 < 4 < 9,$$

the combined estimator has lower variance than either sensor alone.

Solution – Task 3: Role of Covariance

If $\text{Cov}(X, Y) = 0$, then

$$\text{Var}(Z_w) = 4w^2 + 9(1 - w)^2 = 13w^2 - 18w + 9.$$

The optimal weight satisfies

$$26w - 18 = 0, \quad \boxed{w^* = \frac{9}{13}}.$$

The minimum variance is

$$\text{Var}(Z_{w^*}) = \boxed{\frac{36}{13} \approx 2.77}.$$

Since $2.77 < 3.86$, combining the sensors becomes **more useful** when their uncertainty is less correlated.

Covariance measures how much uncertainty is shared across measurements.

Optional Challenge: Zero Covariance but Dependence

Task type: Optional Challenge

Suggested time: 8 minutes

Let $X \in \{-1, 0, 1\}$ with equal probability, and define

$$Y = X^2.$$

You may use

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

1. Compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, and $\mathbb{E}[XY]$.
2. Compute $\text{Cov}(X, Y)$.
3. Determine whether X and Y are independent.
4. Explain why covariance fails to reveal their relationship.

Solution – Optional Challenge: Covariance

Since X takes $-1, 0, 1$ with equal probability,

$$\mathbb{E}[X] = \frac{-1 + 0 + 1}{3} = 0.$$

Since $Y = X^2$,

$$\mathbb{E}[Y] = \frac{1 + 0 + 1}{3} = \frac{2}{3}.$$

Also $XY = X^3$, so

$$\mathbb{E}[XY] = \frac{-1 + 0 + 1}{3} = 0.$$

Therefore,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

Solution – Optional Challenge: Independence

Although

$$\text{Cov}(X, Y) = 0,$$

the variables are not independent because $Y = X^2$.

For example,

$$\mathbb{P}(Y = 0) = \frac{1}{3},$$

but

$$\mathbb{P}(Y = 0 \mid X = 0) = 1.$$

Since

$$\mathbb{P}(Y = 0 \mid X = 0) \neq \mathbb{P}(Y = 0),$$

X and Y are **not independent**.

Zero covariance means no linear co-movement; it does not imply independence.

Tutorial 1: model, derive, calculate, and interpret

Questions and discussion